Programmers as Code Health Guardians

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THE CODE HEALTH GUARDIAN

CODE

REASONABLY SIMPLE

PRAGMATICALLY MAINTAINABLE

HUMAN ROLE

CODE-HEALTH GUARDIANS

A PRODUCTIVITY QUESTION

MINDSET OVER CODE

> The user problem is bigger than the engineering problem

OWNERSHIP MENTALITY

- Engineers as product owners
- Health starts outside the code

THREE PROBLEMS OF COMPLEXITY

- Symptoms that degrade software systems
- Drag down team velocity
- Three varying levels of severity

CHANGE AMPLIFICATION

- Small intended change
- Spreads across hundreds of lines of code
- Affects multiple files
- Least severe problem

COGNITIVE LOAD

- High mental effort required
- Hard to digest code logic
- Hours spent understanding a single function

UNKNOWN UNKNOWNS

- **The most severe problem**
- Developers lack critical info
- Don't know what is missing

CAUSES OF COMPLEXITY

Obscurity

- Poor naming
- Insufficient documentation
- “Clever” code

Complex / Shallow interfaces

- Modules exposing too much internal logic
- Difficult to understand or use

Unfamiliarity / Inconsistency

- Using different solutions to the same problem (“Access checkers”)
- Unfamiliar patterns

DESIGN PRINCIPLES

Deep modules

- Simple interface that hide complex functionality

Design by contract

- Define necessary preconditions
- Guarantee postconditions

Don't repeat yourself (DRY)

- Prevents fixing one instance while leaving others broken
- Link related parts (comments, shared tests)

STRATEGIC PROCESSES

Code health

- Code health doesn’t start when you write code, it starts before.

Design Docs

- Documents used for having our ideas, with their pros and cons

One-pager

- Simple paper where you think through the trade-offs.
- ¿Right solution?
- Fixing a bad idea in paper cost 5 minutes.

FOUR GOALS OF CODE REVIEW

Quality

- Having a second pair of eyes reviewing the code catches things you missed.

Knowledge Sharing

- Over time, code revisions helps all the team understand the whole system.

Delivery Speed

- Optimize flow rather than perfection.

Psychological Safety

- Every member feels safe to ask, admit mistakes, ...